

Trader Performance vs Market Sentiment

Primetrade.ai Round-0 Task | June 2026

Executive Summary

Across 32 traders and 211,224 trades on Hyperliquid (May 2023 – May 2025), merged with daily Crypto Fear & Greed Index data, three findings emerge:

- 1. **Traders earn 2.7× more per trade during Fear than during Extreme Greed** (\$126 vs \$46). The Mann-Whitney U test confirms this is highly significant ($p < 0.001$).
- 2. **The top 5 traders by total PnL are aggressively contrarian** — 97% long during Extreme Fear, only 12% long during Extreme Greed. The bottom 5 are perma-short across all regimes and lose money specifically when sentiment is greedy.
- 3. **Daily total PnL is negatively correlated with the Fear & Greed value** (Spearman $\rho = -0.17$, $p < 0.001$) — as the crowd gets greedier, profits shrink.

Headline takeaway: Among the 32 traders studied, performance dispersion is dominated by directional bias conditioned on sentiment. A simple contrarian framework — long during fear, short during extreme greed — matches the historical winners' pattern.

1. Data Overview

Dataset	Rows	Time Range	Unique Entities
Hyperliquid trader data	211,224	May 2023 – May 2025	32 accounts, 246 coins
Crypto Fear & Greed Index	2,644	Feb 2018 – May 2025	5 sentiment buckets

Merge: Performed on date. Only 6 of 211,224 trades (0.003%) failed to match a sentiment record — dropped without consequence.

Data quality notes: - The task description specified a `leverage` column. **The provided CSV does not contain leverage data.** All leverage-conditioned analysis is therefore excluded. - The `event` column referenced in the task corresponds to `Direction` in the CSV (Open/Close Long/Short). - `Closed PnL` is populated only on close events; open events have `PnL = 0` by design. - 32 traders is a small, curated sample — findings describe these specific traders, not the broader market.

2. Methodology

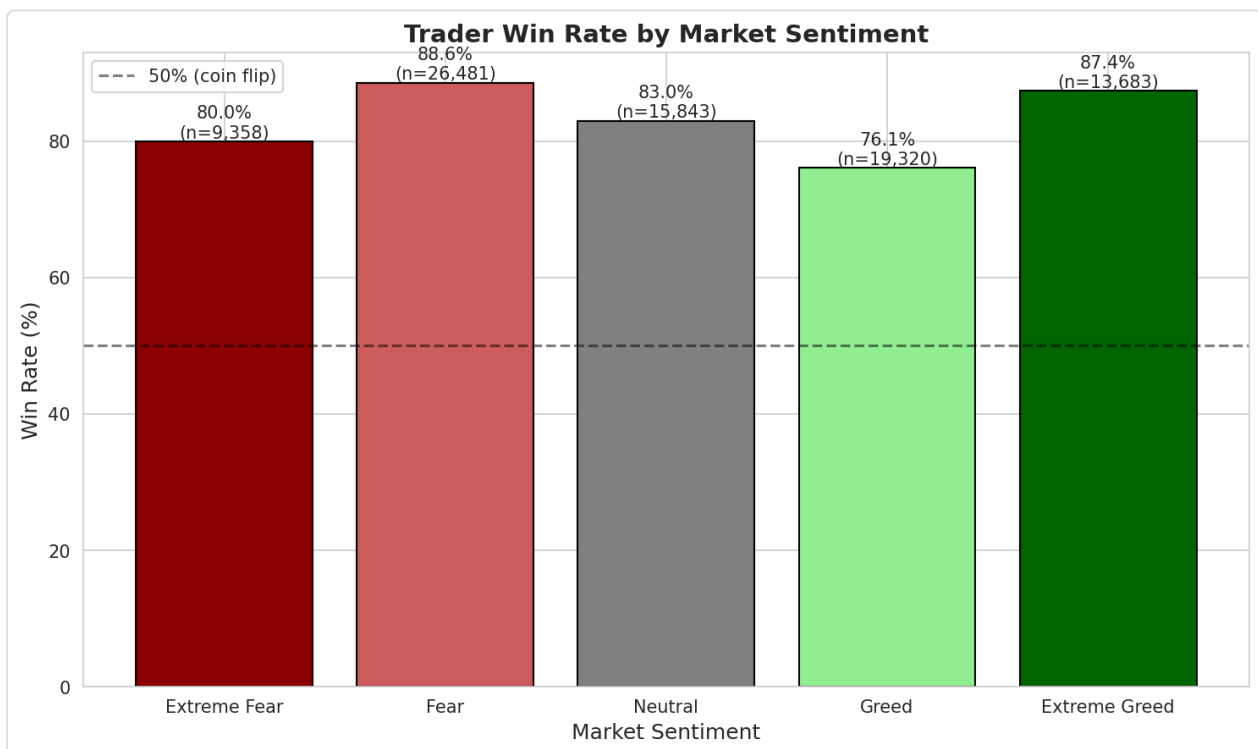
The analysis follows four steps:

1. **Load** both CSVs into pandas, parse timestamps (DD-MM-YYYY HH:MM format for trades, ISO date for sentiment).
2. **Merge** trades to sentiment on date, attaching each trade's daily classification and numeric F&G value.
3. **Segment** trades into close events (for PnL analysis) and open events (for directional bias analysis).
4. **Aggregate and test** — group by sentiment, compute summary stats, run Mann-Whitney U for Fear-vs-Greed PnL difference, and Pearson/Spearman correlation for daily PnL vs F&G value.

All charts are reproducible from `analysis.py` or `analysis.ipynb`.

3. Findings

3.1 Win Rate by Sentiment



Sentiment	Win Rate	Trades
Extreme Fear	80.0%	9,358
Fear	88.6%	26,481

Sentiment	Win Rate	Trades
Neutral	83.0%	15,843
Greed	76.1%	19,320
Extreme Greed	87.4%	13,683

Win rate is highest during Fear (88.6%) and Extreme Greed (87.4%), and lowest during plain Greed (76.1%). The U-shape suggests these traders perform best when sentiment is decisively one-sided. Middle-Greed appears to be the worst environment, likely because the directional signal is ambiguous and traders get whipsawed.

3.2 Realized PnL by Sentiment

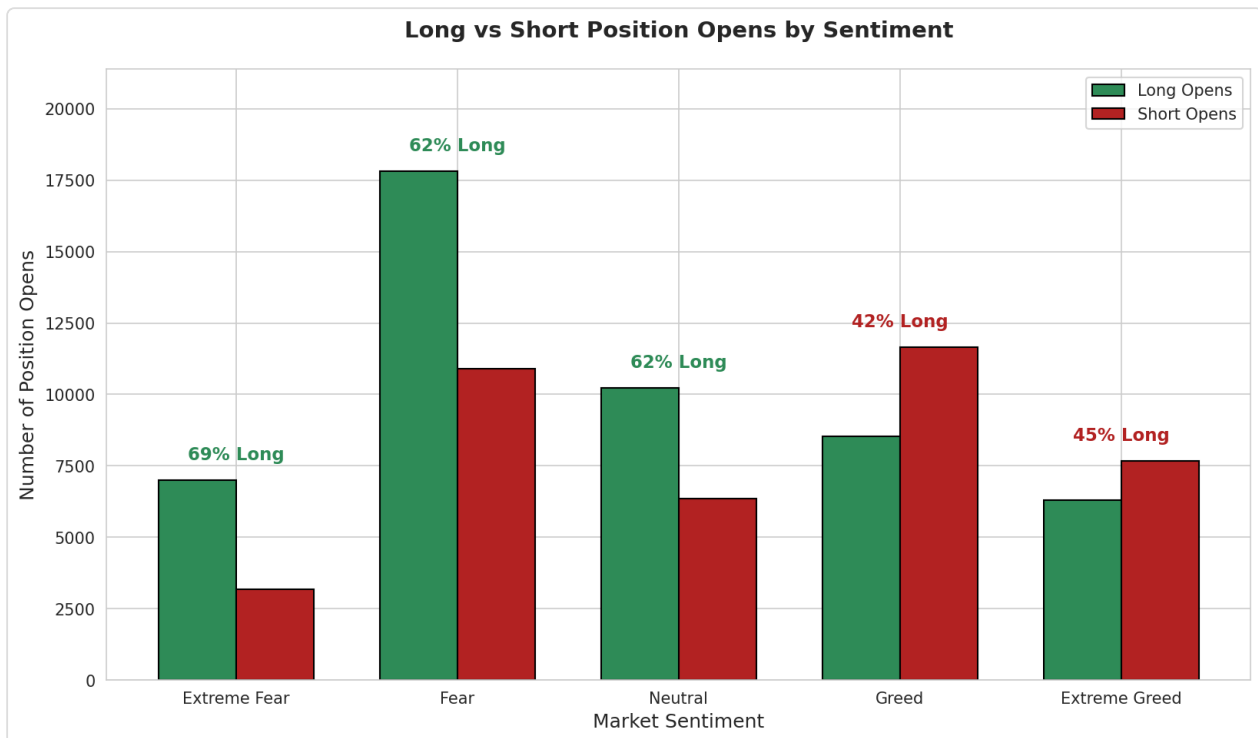


Sentiment	Mean PnL	Median PnL	Total PnL
Extreme Fear	\$95.25	\$8.05	\$891,391
Fear	\$126.41	\$7.11	\$3,347,568
Neutral	\$68.32	\$4.40	\$1,082,347
Greed	\$69.17	\$3.98	\$1,336,414
Extreme Greed	\$46.23	\$7.13	\$632,537

Mann-Whitney U test (Fear days vs Greed days): - Fear days mean: **\$118.28** / Greed days mean: **\$59.66** - U = 635,881,467, **p < 10⁻⁶**

This is the core finding: the most profitable environment for this cohort is moderate Fear, not Greed. Of \$7.3M total profit, \$3.35M (46%) was earned during 'Fear' regimes alone — far more than any other bucket.

3.3 Long vs Short Bias by Sentiment



Sentiment	Long Opens	Short Opens	% Long
Extreme Fear	7,005	3,174	68.8%
Fear	17,824	10,887	62.1%
Neutral	10,222	6,353	61.7%
Greed	8,544	11,664	42.3%
Extreme Greed	6,300	7,663	45.1%

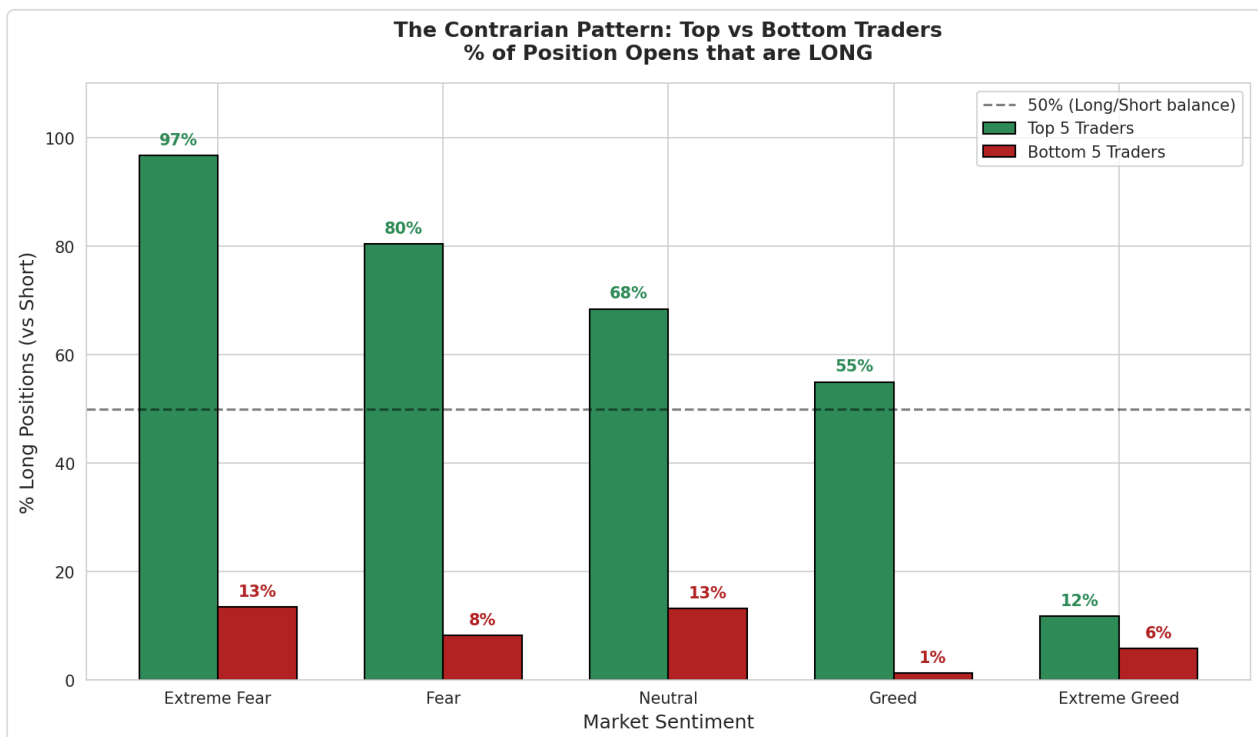
The cohort exhibits a clear contrarian directional bias. As sentiment shifts from Extreme Fear to Greed, the long-bias drops from 69% to 42% — they buy fear and fade greed. The relationship is monotonic.

3.4 Top vs Bottom Traders – The Headline Finding



Trader Group	Total PnL	Trades
Top 5	\$4,613,792	~22,224 closes
All 32 (combined)	\$7,290,257	84,685 closes
Bottom 5	-\$235,818	4,123 closes

The top 5 traders earned more than the remaining 27 traders combined. But **what** they do differently is the interesting part:



Sentiment	Top 5 % Long	Bottom 5 % Long
Extreme Fear	97%	13%
Fear	80%	8%
Neutral	68%	13%
Greed	55%	1%
Extreme Greed	12%	6%

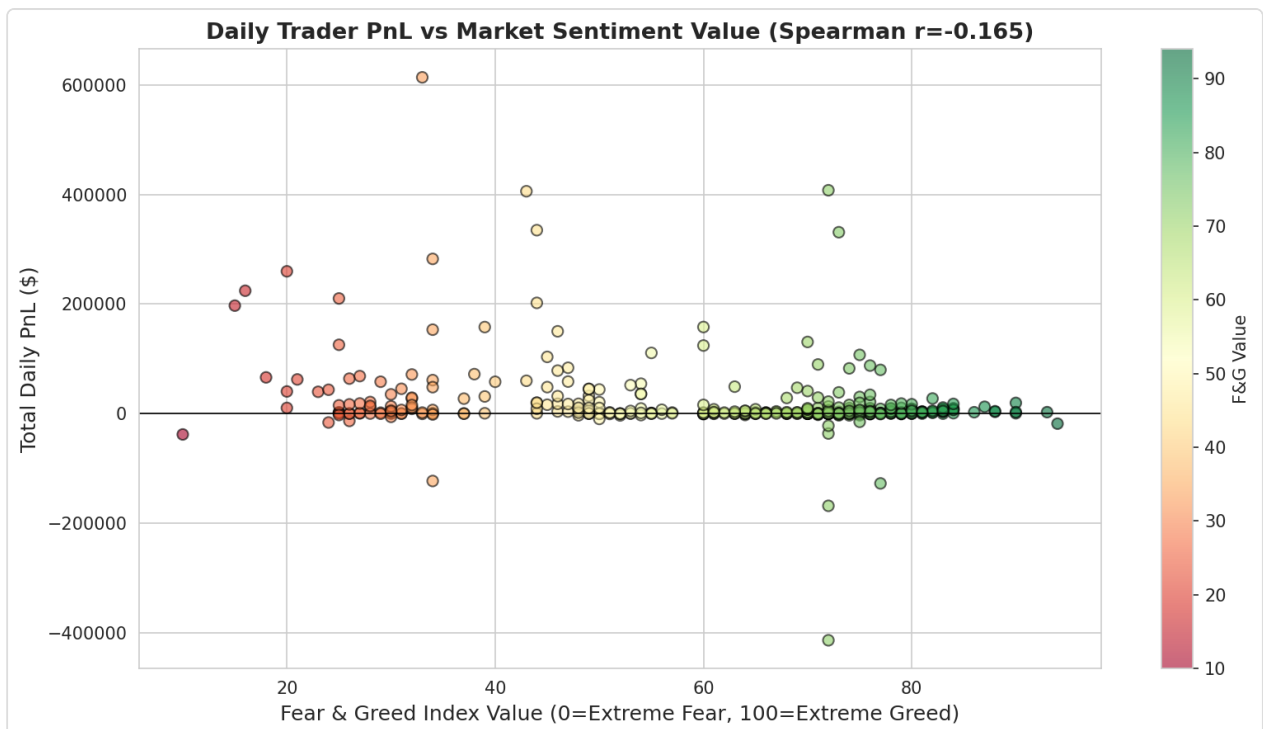
The top traders go aggressively long during Extreme Fear (97%) and aggressively short during Extreme Greed (only 12% long). The bottom traders are essentially perma-short — they bet against the rebounds and got crushed.

The PnL impact is dramatic: - **Top 5 average PnL during Extreme Greed: +\$39 per trade** - **Bottom 5 average PnL during Extreme Greed: −\$568 per trade**

The bottom 5 lose specifically during Greed regimes (avg −\$281 in Greed, −\$568 in Extreme Greed) because they are betting against the market's momentum while it is most directional.

This is the highest-signal finding in the dataset: **position direction conditioned on sentiment explains most of the performance dispersion.**

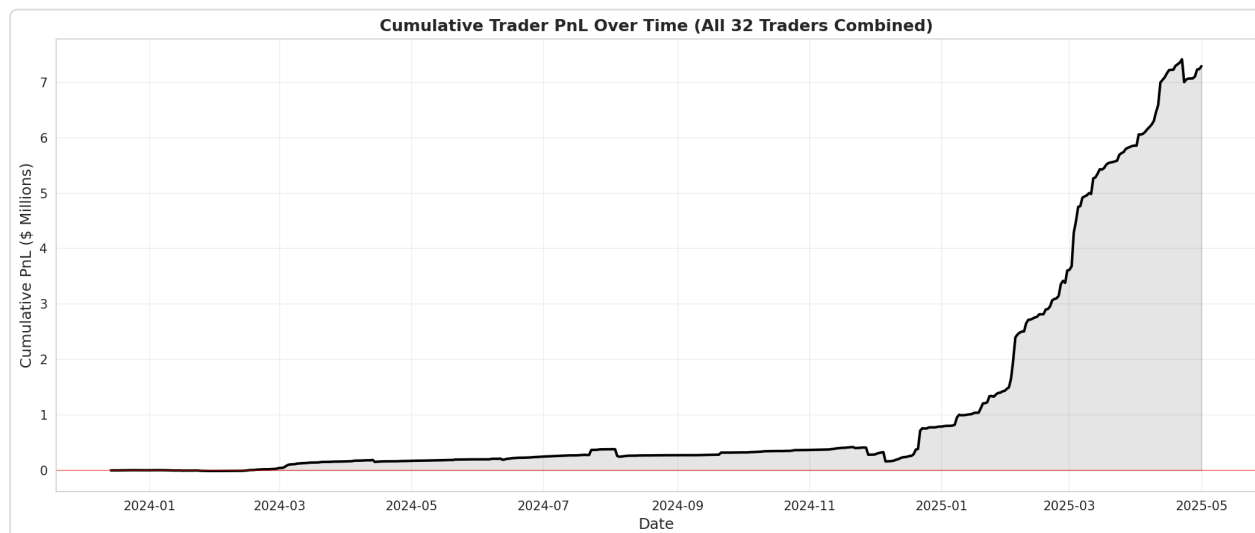
3.5 Daily PnL vs Sentiment Score (Continuous View)



- **Pearson correlation:** $r = -0.250$, $p < 10^{-6}$
- **Spearman correlation:** $\rho = -0.165$, $p = 0.0008$

The correlation is small in magnitude but statistically robust and consistent with every other finding: as the Fear & Greed value rises (market gets greedier), daily aggregate PnL falls.

3.6 Cumulative PnL Over Time



The cohort's cumulative PnL grows steadily over the two-year period to \$7.3M, with no extended drawdowns. This is consistent with a strategy that generates positive expectancy across regimes — though, as shown above, the bulk of the gains come from Fear regimes.

4. Trading Strategy Implications

1. **The cohort's edge is contrarian, concentrated in Fear regimes.** A strategy that goes long during fear and short during greed (especially extreme greed) matches the winners' historical pattern.
2. **Greed periods are the danger zone for unsophisticated traders.** The bottom 5 traders lose virtually all their money during Greed and Extreme Greed regimes. Risk controls (smaller position size, tighter stops) should escalate as sentiment approaches greed extremes.
3. **Direction is more predictive than entry timing.** The dispersion between top and bottom traders is explained almost entirely by *which way* they were betting, not *when*. A simple rule — "don't be short during Extreme Fear, don't be long during Extreme Greed" — would have lifted several bottom-5 traders out of the loss column.
4. **The middle-Greed bucket is uniquely treacherous** (lowest win rate at 76%, lowest mean PnL at \$69). Direction is ambiguous and traders get whipsawed. Position-sizing models could weight signals lower in this regime.

5. Limitations

- **Causation vs correlation:** The Fear & Greed value is partly a function of recent price action. We cannot claim sentiment *causes* trader outcomes — only that they co-move. Endogeneity is a real concern.
 - **Sample size:** 32 traders is a small, curated set. Findings may not generalize to the broader retail population.
 - **No leverage data:** The task description mentioned leverage, but it is absent from the CSV. Leverage-conditioned analyses (e.g., liquidation rates by sentiment) are not possible with the data provided.
 - **Survivorship/selection effects:** Unknown whether the 32 accounts were selected for notability. Caveat all findings accordingly.
 - **No regime-transition handling:** Sentiment buckets are point-in-time. Whipsaw behavior around regime transitions is not modeled and may matter for execution.
 - **No volatility regime control:** Some of the Fear-vs-Greed performance gap may be driven by volatility differences (crypto bear markets have different volatility structure than bulls), which we did not isolate.
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6. Conclusion

The single most important finding in this dataset is that **trader performance dispersion is dominated by directional bias conditioned on sentiment**. Top traders fade the crowd: long during Fear, short during Greed. Bottom traders are perma-short and bleed during rallies. The Fear regime is where most of the profit lives (\$3.35M of \$7.3M total — 46% from 29% of trades).

The simplest actionable rule for this cohort is also the strongest: **be long during Fear, neutral-to-short during Greed, and especially cautious during Extreme Greed where the worst outcomes cluster**.

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